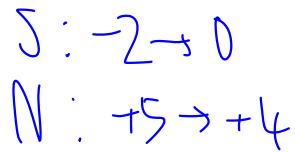
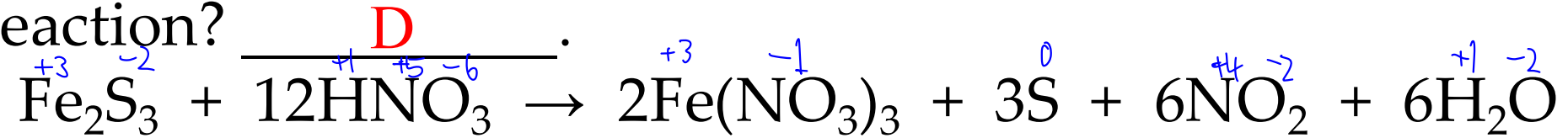


Chapter 20

Electrochemistry

1. Which substance is the oxidizing agent in the following reaction?



A) S

B) NO₂

C) Fe₂S₃

D) HNO₃

E) H₂O

2. What is the oxidation number of manganese in the MnO_4^- ion? E.

A) +1

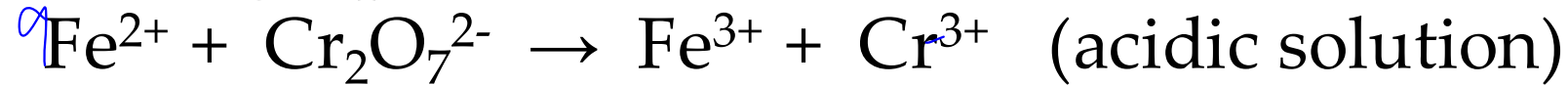
B) +2

C) +5

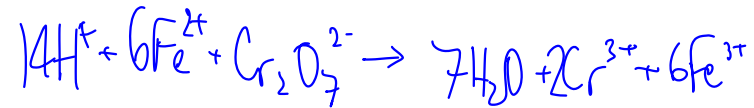
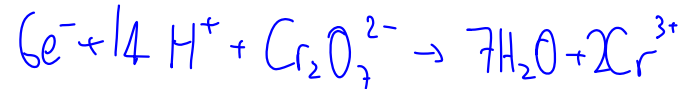
D) +4

E) +7

3. What is the coefficient of the dichromate ion when the following equation is balanced?



 A .



A) 1

B) 2

C) 3

D) 5

E) 6

4. The purpose of the salt bridge in an electrochemical cell is to E.

A) provide a source of ions to react at the anode and cathode ✕

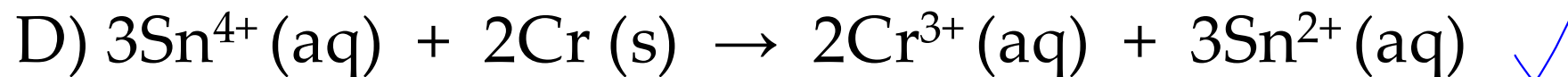
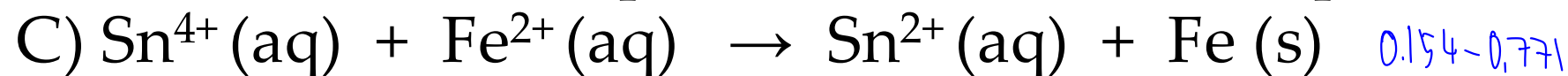
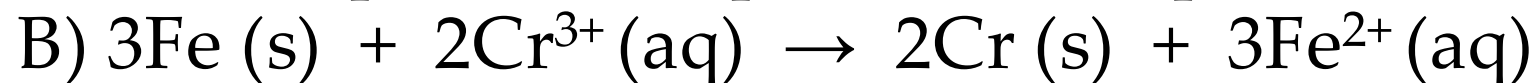
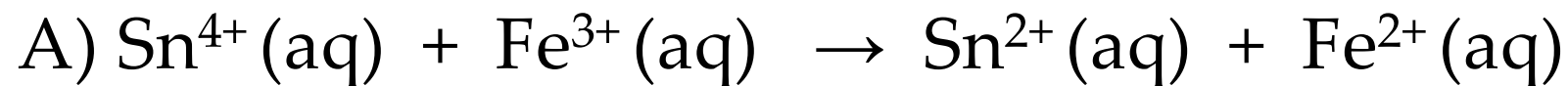
B) Provide oxygen to facilitate oxidation at the anode ✕

C) provide a means for electrons to travel from the anode to the cathode ✕

D) provide a means for electrons to travel from the cathode to the anode

E) maintain electrical neutrality in the half-cells via migration of ions

5. Which of the following reactions will occur spontaneously as written? D .



Half-reaction	E° (V)
$\text{Cr}^{3+}(\text{aq}) + 3\text{e}^- \rightarrow \text{Cr}(\text{s})$	-0.74
$\text{Fe}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Fe}(\text{s})$	-0.440
$\text{Fe}^{3+}(\text{aq}) + \text{e}^- \rightarrow \text{Fe}^{2+}(\text{s})$	+0.771
$\text{Sn}^{4+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Sn}^{2+}(\text{aq})$	+0.154

6. Consider an electrochemical cell based on the reaction:

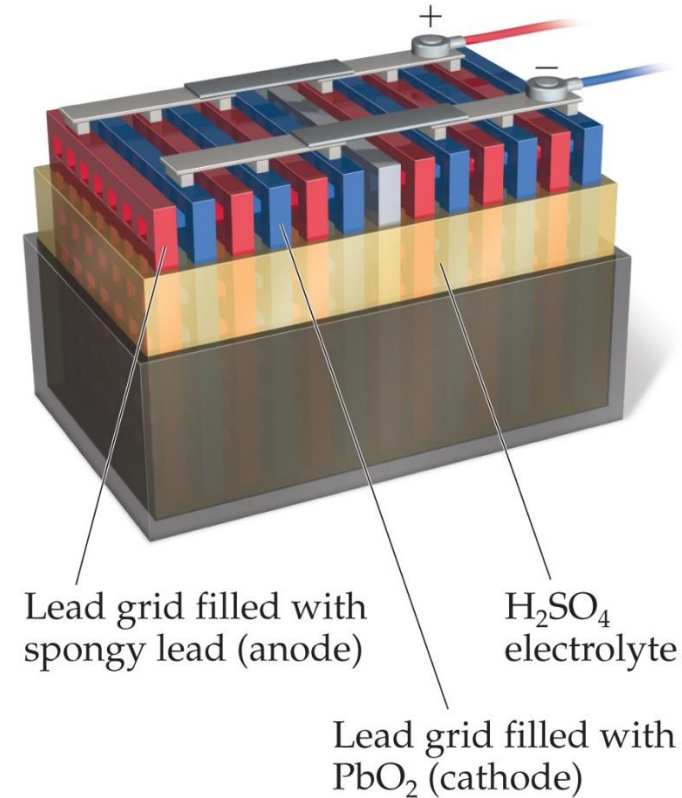


Which of the following actions would not change the measured cell potential? B.

- A) lowering the pH in the cathode compartment? [H⁺] change
- B) addition of more tin metal to the anode compartment** Intensive
- C) increasing the tin (II) ion concentration in the anode compartment ✓
- D) increasing the pressure of hydrogen gas in the cathode compartment
- E) Any of the above will change the measured cell potential

7. In a lead-acid battery, the electrodes are consumed. In this battery, A.

- A) the anode is Pb
- B) the anode is PbSO_4
- C) the anode is PbO_2
- D) the cathode is PbSO_4
- E) the cathode is Pb



8. Cathodic protection of a metal pipe against corrosion usually entails C.

- A) attaching an active metal to make the pipe the anode in an electrochemical cell
- B) coating the pipe with another metal whose standard reduction potential is less negative than that of the pipe
- ☒ C) attaching an active metal to make the pipe the cathode in an electrochemical cell
- D) attaching a dry cell to reduce any metal ions which might be formed
- E) coating the pipe with a fluoropolymer to act as a source of fluoride ion (since the latter is so hard to oxidize)

9. One of the differences between a voltaic cell and an electrolytic cell is that in an ^{electrolysis} electrolytic cell, C.

- A) an electric current is produced by a chemical reaction
- B) electrons flow toward the anode
- C) a nonspontaneous reaction is forced to occur
- D) O₂ gas is produced at the cathode
- E) oxidation occurs at the cathode